Original Article

Optical and Nanostructural Characterization of ZnO Thin Films Prepared by Spray Pyrolysis: Combined Experimental and DFT-GGA Investigation

Mostefa Benhaliliba*

Film Device Fabrication–Characterization and Application FDFCA Research Group USTOMB University, 31130, Oran, Algeria

Use your device to scan and read the article online

Citation Mostefa Benhaliliba, **Optical and Nanostructural Characterization of ZnO Thin Films Prepared by Spray Pyrolysis: Combined Experimental and DFT-GGA Investigation.** *Eurasian J. Sci. Technol.*, **2026**, *6*(3), 212-229.

https://doi.org/10.48309/ejst.2026.561329.1287

CC BY

Article info:
Received: 2025-11-21
Accepted: 2026-03-06
Available Online: 2026-05-21
ID: EJST-2511-1287

Keywords:
ZnO; Spray pyrolysis; Transmittance; Nanostructures; X-ray-XPS; GGA computational

ABSTRACT

Using a combination of experimental and density functional theory (DFT) methods, zinc oxide (ZnO) thin films were effectively formed on glass substrates by spray pyrolysis at 250 °C. Atomic force microscopy (AFM), photoluminescence (PL) spectroscopy with He–Cd laser excitation, UV–Vis spectroscopy, X ray analysis and X-ray photoelectron spectroscopy (XPS) study were used to investigate the optical, morphological and crystalline structure characteristics. ZnO's wide-band-gap semiconducting nature and suitability for optoelectronic applications are confirmed by the films' remarkable optical transparency in the visible spectrum and clearly defined optical band gap. The ZnO lattice's intrinsic defect states are responsible for the prominent blue-green emission bands found in PL analysis. The homogenous nanostructured surface morphology revealed by AFM studies emphasizes the significant impact of the deposition procedure on the formation of nanoscale films. The energy stability of ZnO in the wurtzite (Wz) crystal structure is confirmed by complementary DFT calculations carried out under the generalized gradient approximation (GGA), which also offer theoretical justification for the optical behavior seen in experiments. The efficacy of spray pyrolysis for creating superior ZnO thin films with promising optical and nanostructural properties is demonstrated by the great match between experimental results and theoretical expectations.

Introduction

ZnO has attracted significant research interest over the past five decades due to its multifunctional nature and wide range of technological applications. In its bulk form, ZnO is a white, water-insoluble powder that is chemically stable under normal conditions. It is an intrinsic n-type semiconductor with a wide direct band gap of approximately 3.37 eV. ZnO crystallizes primarily in the hexagonal Wz structure, although zinc blende (Z_B) and rock-salt phases may form under specific growth conditions or high pressure. Despite strong competition from materials such as silicon (Si), gallium arsenide (GaAs), and perovskites, ZnO continues to be a promising candidate for next-generation devices due to its unique physical and chemical properties [1–3].

*Corresponding Author: mbenhaliliba@gmail.com (M. Benhaliliba)

Owing to its excellent optical, electrical, and piezoelectric characteristics, ZnO has been widely explored for applications in solar cells, optoelectronic devices, thin-film transistors, gas and chemical sensors, as well as energy storage and conversion systems [4–8]. Furthermore, ZnO exhibits strong coupling with surface acoustic wave (SAW) devices, enhancing its functionality in sensing and signal processing applications [9]. Beyond electronics, ZnO has gained attention in biomedical and environmental fields, where it demonstrates photocatalytic activity, antibacterial behavior, self-cleaning capability, and compatibility with biomedical devices, as summarized in Table 1 [10–12].

Table 1: The different applications of ZnO thin film

Devices	Applications	[Ref.]
Solar cell	1/Window layer or buffer layer in: -Dye-sensitized solar cells -Perovskite solar cells -Cu (In, Ga) Se_2 thin-film solar cells 2/Acts as a transparent electrode and anti-reflective coating to enhance light absorption	[4]
Optoelectronic devices	1/Transparent Conducting Oxides (TCOs): -ZnO thin films can replace indium tin oxide (ITO) as transparent electrodes in solar cells and displays. -Used in LEDs, LCDs, OLEDs, and touchscreens. 2/Photodetectors & UV Sensors: -Due to its wide bandgap, ZnO is highly sensitive to UV light. -Used in UV photodiodes, UV detectors, and solar-blind photodetectors. 3/Light-Emitting Devices: -Can be used in UV and blue light-emitting diodes (LEDs) and laser diodes.	[5]
Gas and chemical sensors	1/ZnO thin films exhibit changes in electrical resistance when exposed to gases. 2/Used for detecting: -Ethanol, hydrogen, ammonia, CO, NO_2, O_2, etc. 3/Applications: Environmental monitoring, automotive sensors, industrial safety systems.	[6]
Energy storage and conversion devices	1/Supercapacitors: ZnO thin films act as electrode materials due to high surface area and good conductivity. 2/Batteries: Used as anode materials in Li-ion and Na-ion batteries. 3/Piezoelectric generators: ZnO thin films exhibit piezoelectric properties, enabling nanogenerators and pressure sensors.	[7]
Thin-film transistors (TFTs)	1/ZnO thin films are used as the semiconducting layer in TFTs. 2/Applications: Flat-panel displays, e-paper, and flexible electronics	[8]
Photocatalysis and environmental applications	1/ZnO thin films can degrade organic pollutants under UV irradiation. 2/Used in: -Photocatalytic degradation of dyes and pollutants -Water and air purification	[10]
Antibacterial and self-cleaning coatings	1/ZnO has antimicrobial properties and photocatalytic activity under UV light. 2/Used in: -Self-cleaning glass -Antibacterial coatings for medical devices -Water purification systems	[11]
Biomedical applications	1/Biosensors: ZnO's biocompatibility and surface reactivity enable enzyme or DNA immobilization.	[12]

	2/Drug delivery and bioimaging applications due to its nontoxic and semiconducting nature.	
SAW devices	ZnO thin films' piezoelectric properties make them suitable for: -Filters -Oscillators -Resonators in communication systems.	[9]

A variety of techniques have been developed for the fabrication of ZnO thin films, including spray pyrolysis, spin coating, radio-frequency (RF) sputtering, and hydrothermal methods, each offering distinct advantages and limitations [13-16]. Among these, spray pyrolysis stands out as a low-cost, simple, and scalable technique that does not require vacuum conditions, making it suitable for large-area coatings.

In this work, ZnO thin films are deposited using a facile and economical spray pyrolysis method. The optical properties of the films are investigated using UV–Vis–IR spectroscopy and PL analysis, while surface morphology is examined to assess film uniformity and nanostructure. Additionally, DFT calculations employing the GGA are used to support and interpret the experimental findings.

The novelty of this work lies in the combined experimental–theoretical investigation of ZnO thin films deposited by a low-cost, vacuum-free spray pyrolysis technique. Unlike many studies that focus solely on fabrication or characterization, this research correlates optical transparency, PL emission related to oxygen vacancies, and uniform nanostructured morphology with first-principles DFT calculations using the GGA approximation. The study highlights the role of intrinsic defects in governing optical emission and demonstrates the suitability of spray-deposited ZnO films for multifunctional applications such as dye-sensitized solar cells, piezoelectric devices, and gas sensors. Additionally, the comparative analysis of structural stability and equilibrium volume (V_0) provides insight into the influence of computational approximations on ZnO crystal optimization.

Experimental

This sample of ZnO based thin film is produced by spray pyrolysis at 250 °C onto glass substrate as shown in Figure 1. Others famous techniques like spin coating, RF Sputtering, hydrothermal and spray pyrolysis are also reported as portrayed in Figure 2 [13-16].

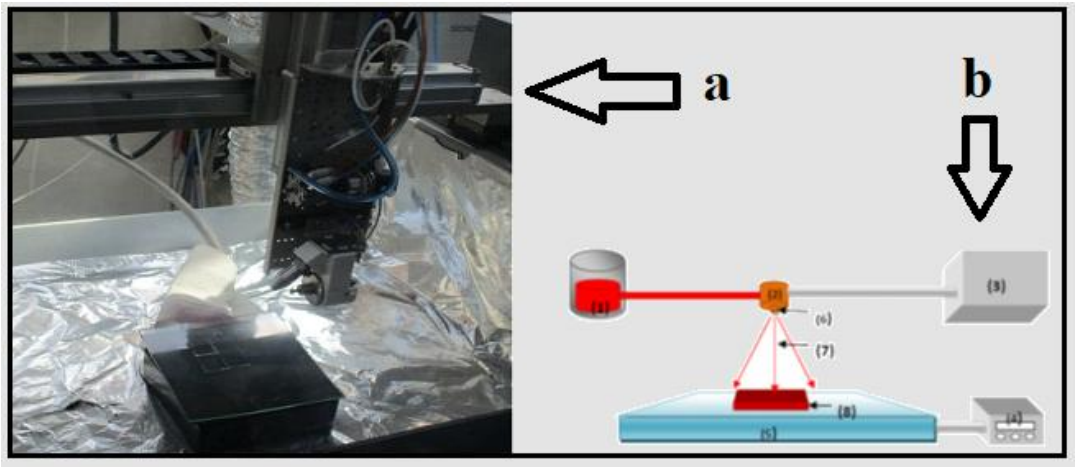


Figure 1: Spray pyrolysis set-up (a), operational schematic of spray pyrolysis (b), flow control (1), atomizer (2), reservoir of solution (2), heater and temperature control by thermocouple (4), substrate holder (5), nozzle (6), droplets (7), substrate (8)



Figure 2: Different techniques to produce ZnO thin films: (1) Spin coating, (2) RF sputtering, (3) Hydrothermal and (4) spray pyrolysis [13-16]

The starting material, 0.05 Mole of zinc acetate with 2 moles of water $Zn(C_2H_3O_2)_2.2H_2O$, is dissolved in methanol. After cleaning process of substrate, the prepared solution is sprayed onto substrate at 250 ºC during 10 min, the distance nozzle-substrate is fixed at 5 cm and spray rate at 1 ml/min. A digital thermometer attached to the heater regulates the glass substrate that is heated to 250 °C. The UV-VIS-IR Shimadzu UV-3101PC double beam spectrophotometer is used to evaluate the optical parameters, and the PAN analytical 790CACX' Pert PRO diffractometer is used to measure the X-ray pattern of as-grown films. Surface texture is examined by atomic force microscope AFM Quesant model 250. A 325 nm, 15 mW He-Cd laser is used in an experimental setup for room temperature (RT) PL study. At a fixed scan rate of 2 Hz, the resolution for the square pictures (3µm × 3µm) was 300×300 pixels. The WSXM system software is used to determine the analyzed AFM parameters.

Results and Discussions

UV-Vis-IR Measurements

Transmittance T (%) of the sputtered ZnO thin film against incident photon wavelength λ(nm) is plotted within UV-Vis-IR (300-900) nm range as seen in Figure 3. An increase of transmittance from UV is recorded reaching two consecutive peaks at 456 and 649 nm. These peaks are determined at 2.73 and 1.91 eV using 1240/λ(nm)=E(eV). Such oscillations are optical interferences produced by various reflections inside thin film. It is due to smoothness and transparency of film due also to film thickness, surface roughness of film and non-uniformity of layer [1, 2]. Furthermore, ZnO is direct transition band gap semiconductor, Eg is extracted using the Tauc plot (not shown here) expressed as [1,2, 13];

$$(\alpha h\nu)^2 = A(h\nu - Eg) \quad (1)$$

Where α is the absorption coefficient, h is the planck constant, ν is the photon frequency and A is the proportionality constant.

Another method to extract the band gap value is to plot the dT/dλ versus λ as seen in Figure 4. This latter shows a peak at 3.25 eV as assigned by an arrow. Figure 4 depicts the behavior of such derivatives (dT/dλ) versus photon energy for the sample in a wide energy range, which includes the band-gap edge [20].

Localized energy states are created within the band gap and close to the band edges in semiconductor crystals when various imperfections, such as impurity atoms,

vacancies, etc., change the lattice homogeneity. For smaller band-gap photon energies, the absorption coefficient does not change abruptly. Instead, it increases gradually with energy. This increase follows an exponential trend. The behavior is related to the perfection of the crystal lattice and is known as the Urbach tail [20]. Using these consecutive peaks, we determine the layer thickness (d) according to [21];

$$d = \frac{\lambda_1 \lambda_2}{2n(\lambda_1 - \lambda_2)} \quad (2)$$

where n is refractive index about 2 in visible [1]. The thickness of ZnO film is found to be 383.15 nm. Moreover, the optical band gap is found to be 3.25 eV as listed in Table 2. This finding confirms that ZnO is a wide band gap (Eg) semiconductor compared to Si (1.12 eV), GaAs (1.42 eV) or GaN (3.40 eV). Eg of ZnO is around 3.25-3.37 eV, similar values are previously mentioned [17-19].

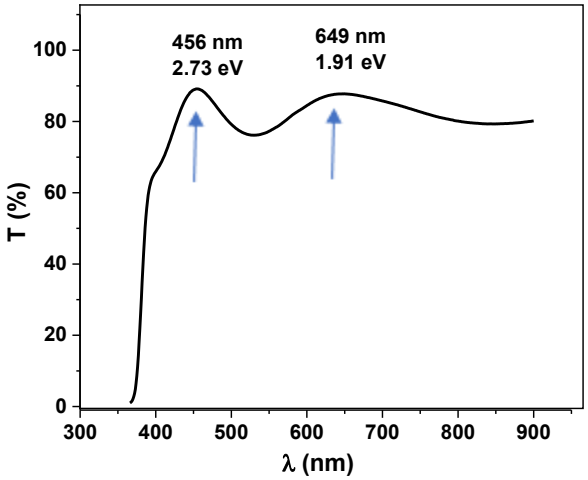


Figure 3: Transmittance-wavelength variation of ZnO film sprayed at 250 ºC. Oscillations are showed by arrows

Table 2: Comparison between parameters of ZnO and other most used semiconductors

Material	ZnO current work	Si [17]	GaAs [18]	GaN [19]
Bandgap (eV)	3.25- 3.37	1.12	1.42	3.40
Exciton binding energy (meV)	60	~15	4.2 at 0K	~20-25 at 77 K and 300 K

PL Spectrum by He-Cd Laser

The PL emission spectra at ambient-temperature of our sample are measured in the 350–600 nm wavelength range. The sprayed films were irradiated at RT with a 15mW He-Cd laser with a wavelength of 325 nm (3.81 eV), and the resulting spectrum is plotted as a response. Figure 5 shows how the PL of sprayed ZnO at different temperatures depends on photon energy within the 2–3.5 eV energy range. As seen in Figure 5, ZnO films show intense yellow luminescence at 2.109 eV and additional blue light emission at 2.805 eV. Intense emissions are located at (441.6 nm-2.80 eV), (534.2 nm-2.32 eV), (537.8 nm-2.30) and (588.0 nm-2.11 eV) as portrayed in Figure 5. These findings correspond to blue, green and yellow light emissions respectively. Figure 5 shows that the spectrum lacks a significant UV near band edge emission, which is indicative of a high-quality ZnO thin film. Whereas, it appears an intense peak around 600 nm noted d, caused by defect occurrence related to deep level emission

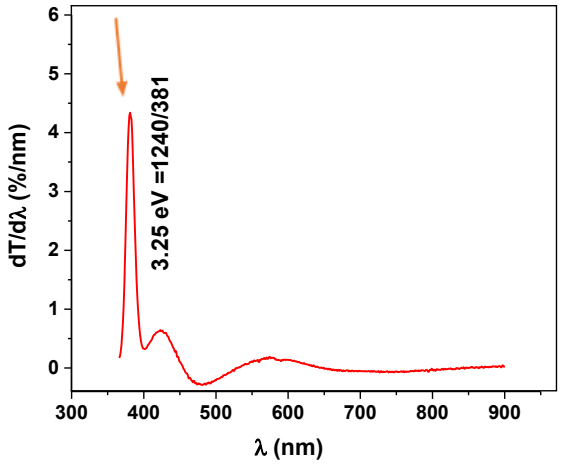


Figure 4: The curve of dT/dλ against photon wavelength of ZnO based thin film. Inset shows the value of bandgap Eg as indicated by arrow.

Previous researches are in well agreement with our findings like Li's work [22]. The PL diagram of ZnO thin films with different Al doping amounts at ambient is previously reported. A green emission peak at 527 nm and a bluish-green peak at 485 nm. The presence of a particular quantity of doping-induced defect states is indicated by the visible peaks at 485 and 527 nm.

The bluish-green (between blue and green) emission at 485 nm is associated with oxygen vacancies, which supply shallow donor levels and promote electron transitions followed by photon emission [22].

Intense emissions are a (441.6 nm-2.80 eV), b (534.2 nm-2.32 eV), c (537.8 nm-2.30) and d (588.0 nm-2.11 eV) as sketched in Figure 5.

Peak by peak are plotted separated as shown in Figure 6 with their Gaussian deconvolution fitting.

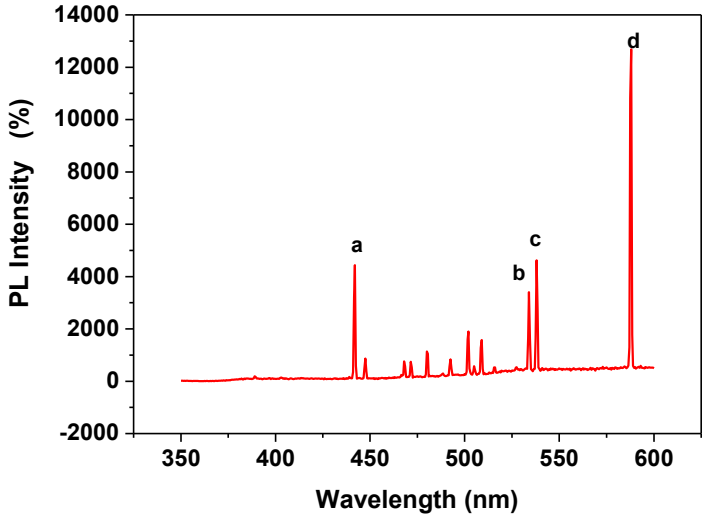
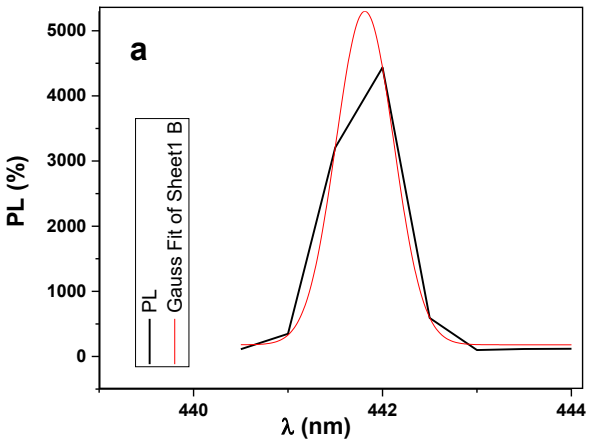
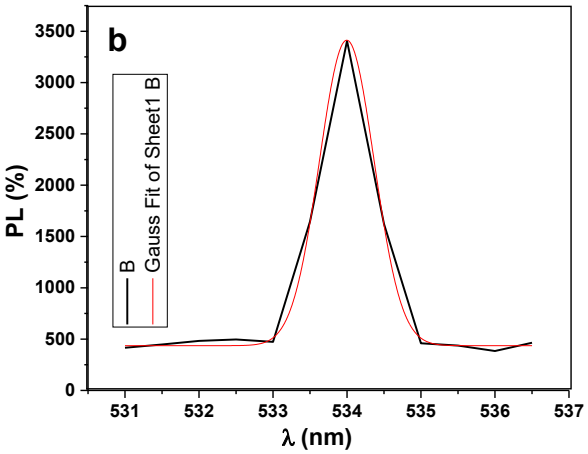
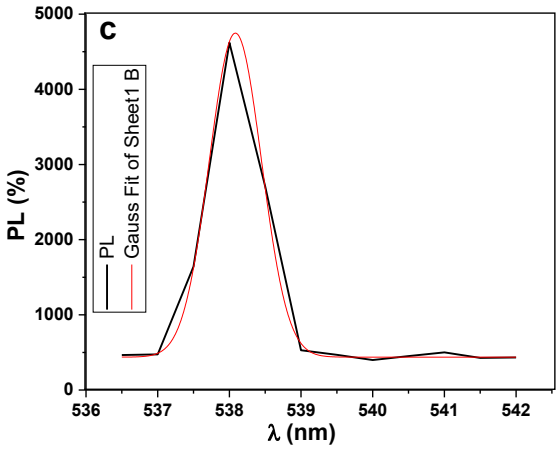


Figure 5: PL spectrum versus photon wavelength of sprayed ZnO layer. The main peaks are a, b, c and d.


a
PL
Gauss Fit of Sheet1 B
PL (%)
λ (nm)
5000
4000
3000
2000
1000
0
440
442
444


b
B
Gauss Fit of Sheet1 B
PL (%)
λ (nm)
3500
3000
2500
2000
1500
1000
500
0
531
532
533
534
535
536
537


c
PL
Gauss Fit of Sheet1 B
PL (%)
λ (nm)
5000
4000
3000
2000
1000
0
536
537
538
539
540
541
542

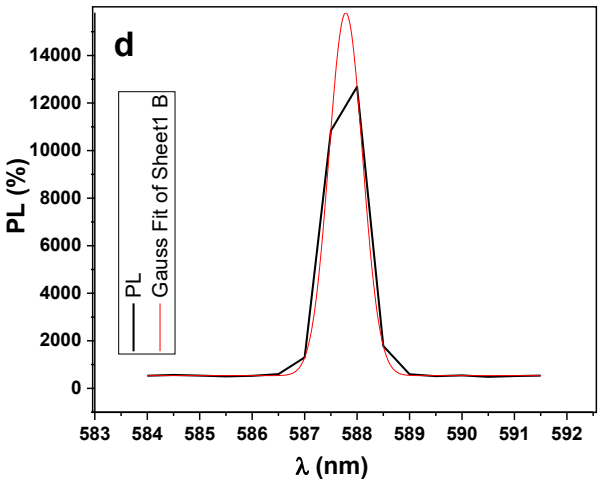


Figure 6: Deconvolution curves by Gaussian fit of the main peaks (a, b, c and d) of PL spectrum versus photon wavelength of sprayed ZnO layer

Parameters of Gaussian fitting of the PL main peaks, are determined by Gaussian expression as follows [23],

$$y = y_0 + \frac{A}{w\sqrt{\frac{\pi}{2}}} e^{-2\left(\frac{x-x_c}{w}\right)^2} \quad (3)$$

Where x_c is the mean (the peak's center), w is the standard deviation (determining the curve's spread). Parameters of Gaussian fitting of PL intense peaks (a, b, c and d) are gathered in Table 3.

Table 3: Parameters of Gaussian fitting of the PL intense peaks (a, b, c and d)

Peaks positions	y_0 (a.u.)	x_c (nm)	W (FWHM)	A Area (a.u.)
A	128.00	432.01	0.62	3968.09
B	416.01	512.02	0.72	2688.10
C	768.02	512.03	0.66	3200.10
D	512.02	576.02	0.62	12288.38

It is commonly known that the near band edge UV range spans roughly 370–390 nm. This range results from exciton recombination near valence and conduction bands (C_B). Within the visible spectrum, deep level emission is found in specific blue, green, yellow, and red wavelengths.

The green emission observed in the 500–550 nm wavelength range, corresponding to photon energies of 2.2–2.5 eV, is attributed to defect states arising from zinc interstitials and oxygen vacancies. The prominent green peaks like b and c shown in Figures 5 and 6 are frequently caused by lack of oxygen, poorer crystalline structure quality, and a high density of defects. Peak D, on the other hand, has energies of 1.9–2.2 eV and is a part of the yellow-orange band. These are caused by deep acceptor levels, OH-related defects, and the oxygen interstitial (O_i) location. The blue range produced by zinc vacancies and donor-acceptor pair transitions is represented by peak A, which lies between 430 and 470 nm. Consequently, visible emissions are strong caused by high defect density.

At RT, the luminescence emission bands are seen at 487 nm (blue emission) and 530 nm (green emission). It is noted that none of the ZnO exhibit PL peak modification positions [24].

Strong PL in the visible-NIR (1.6–2.2 eV) and UV (~3.33 eV) is seen in the as-grown and annealed Al-doped ZnO thin film samples. The broad visible-NIR PL corresponds to the radiative

transition associated with oxygen interstitial defects in the ZnO structure, whereas the UV PL peak originates from the near band edge emission of crystalline ZnO [25].

PL is used to describe the optical characteristics of ZnO thin films following the annealing process. At RT, a mild defect-band emission and a robust free exciton emission were seen in the visible area [26]. The free exciton emission of 67.54 meV has been measured at ambient based on the full-width at half-maximum. Annealing the ZnO film in an oxygen environment increases the PL intensity peaks in the visible light and UV light areas. On the other hand, annealing it in a nitrogen atmosphere selectively increases just the PL intensity peak in the UV light area [26].

AFM Observation

The surface texture of ZnO layer is observed by an atomic force microscope advanced technique. The 3D-Scanned AFM pictures (A), 2D image (B), Histogram delivered by AFM analysis (C) and raw material (D) of ZnO film grown by spray pyrolysis are displayed in Figure 7.

The surface of 3D scanned AFM (3µ×3µ) images looks like dome with z-axis orientation and different sizes as observed in Figure 7A. As observed in surface, the morphology of the film is nearly uniform, and there are no obvious holes or flaws. In 2D-picture (3×3µ) , we see several domains delimited by boundaries as drawn in Figure 7B.

According to Figure 7C, the root mean square, which estimates the roughness of film surface, is about 23-25 nm.

The histogram, which shows the percentage against nanograin height, is shown in Figure 7C. Nonetheless, the statistical distribution's form depicts a height of 35 nm. Such a histogram is observed to resemble a Gaussian distribution. Such analysis reveals the occurrence of nanograins in ZnO thin film and it is in well agreement of previous studies.

The histogram particle statistics is displayed in Figure 7E.

Figure 7A displays the nanoscale surface morphology with roughness, grain distribution and growth texture along z- axis.

Closely packed nanograins are assembled on the surface of sprayed ZnO films. This surface shows mountains and valleys with different grain heights and rough surface texture. It is a granular morphology and peaks appear periodically arranged. This profile corroborates with polycrystalline structure along (002) orientation [1,2]. Moreover, ZnO film exhibits a rough surface depending on the deposition parameters like substrate temperature, precursor amount and annealing process. It is also explained by the occurrence of higher peaks related to larger grains and lower valleys related grain boundaries as seen in Figure 7B.

It is concluded that the ZnO film is uniform but nanostructured having a regular distribution of grains. As shown in Figure 7B, no voids are present.

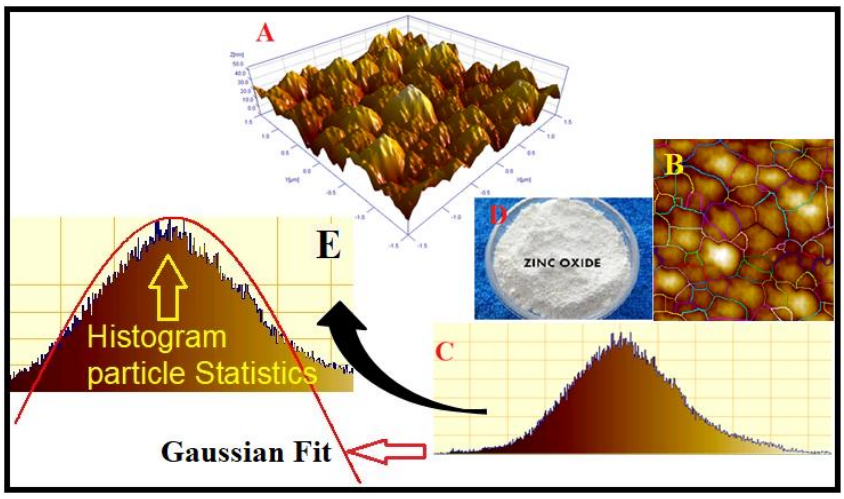


Figure 7: 3D-Scanned AFM pictures (A), 2D image (B) , histogram delivered by AFM analysis (C) and raw material (D), histogram particle statistics (Gaussian fit) (E) of ZnO film grown by spray pyrolysis

As seen in previous works, the size distribution of pure ZnO nanoparticles and Mg-doped ZnO (MZO) nanoparticles is studied. Higher doping concentrations are found to cause an increase in particle size and agglomeration [27]. Analyze the granular (nanostructured) nature of the ZnO films and the consistent, discernible increase in grain size with deposition time (film thickness) [20].

X-Ray Measurements

Figure 8 shows the X-ray pattern of ZnO film within 20°-100° 2θ angle range of recording main peaks (a) 34.200, (b) 36.12°, (c) 47.60° and (d) 63.00°. The JCPDS :36-1451 card indicates the main peak of ZnO at 34.42°, according to obtained results, we can confirm that main peak is (a) as shown in Figure 8, indexed (002). This latter records a shift of 0.22°. Exploring the X-ray spectra, we observe that the pattern displays sharp peaks, which means the film is crystalline and not amorphous structure. X-ray pattern shows that the films are polycrystalline in nature with hexagonal Wz crystal structure. The main orientations (002), (101), (102), and (103) are located respectively at 34.20°, 36.12°, 47.60° and 63° (2θ) angle set.

Thereafter, ZnO normally crystallizes in the hexagonal Wz structure, and the main peaks (a, b, c, d) match well with that phase. The profile of each peak can be interpreted as listed in Table 4.

Table 4: Peak position and their properties from X-ray analysis

Peak	Position (°)	Properties
a	34.20	Indexed as (002) plane of hexagonal ZnO Indicates crystal growth along the a-axis Presence of this intense peak confirms polycrystalline ZnO
b	36.12	Indexed as (101) plane This is an intense peak, which is very important
c	47–48	Indexed as (102) plane Confirms the hexagonal Wz structure Shows that crystal growth is not limited to only one direction
d	62–63	Indexed as (103) plane Further evidence of polycrystalline nature Indicates good lattice formation

Based upon X-ray spectra, we extract the grain size is found to be 39 nm, 94 nm and 221 nm according (002), (101) and (102) orientations respectively. Textural coefficient is 1.56, 0.58 and 0.86 according the main cited orientations respectively.

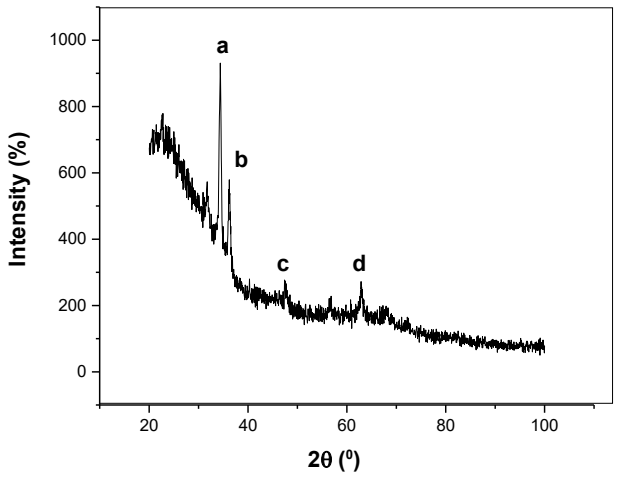


Figure 8: X-ray of ZnO thin film, a (002), b (101), c (102) and d (103) are the main peaks

Phase Purity, Peak Width, and Sharpness

It is revealed that no extra or unidentified peaks are visible, no secondary phases (such as Zn, $Zn(OH)_2$, or ZnO_2), so the film is single-phase ZnO.

Sharp and narrow peaks demonstrate and confirm a good crystallinity. Moreover, the slight peak broadening reveals the nanocrystalline grain size, which is common in thin films. (Using the Scherrer equation, crystallite size can be calculated if peak FWHM is known [2, 3].)

Consequently, ZnO thin film has hexagonal Wz structure, film is polycrystalline with strong c-axis (002) orientation. It is noted that a good crystallinity and a phase purity are suitable for UV photodetectors, transparent conducting layers, gas sensors and piezoelectric devices. Some works reported such results [28].

X-ray pattern confirms that the sprayed ZnO phase is grown along a (002) preferential plane, caused by the lower free surface energy of this plane. These results corroborate with those reported in the literature [1-3]. The Strong (002) peak means the ZnO film is preferentially oriented along the c-axis. This is typical for well-grown ZnO thin films and c-axis orientation is desirable for Optoelectronic devices, piezoelectric applications and sensors

XPS Investigation of ZnO Thin Film

The films' chemical compositions and defects were examined using XPS measurement.

The elemental states, surface chemical composition, and defect-related characteristics of the ZnO thin films were examined using XPS as portrayed in Figure 9. With no discernible impurity-related peaks, the broad survey spectrum validates the existence of zinc and oxygen (O) as the primary ingredients, suggesting good chemical purity of the deposited films. Figure 9a shows the spectrum of XPS of ZnO within 0-1200 eV range. Such plot records three narrow peaks around 1050, 550 and 300 eV.

The Zn $2p_3/_2$ and Zn $2p_1/_2$ are represented by two distinct peaks in the high-resolution Zn 2p spectra, which are centered at binding energies of roughly 1021.5–1022.0 eV and 1044.5–1045.0 eV, respectively as seen in Figure 9b. The presence of zinc in its fully oxidized form is confirmed by the energy separation between these two peaks (~23 eV), which is typical of the Zn^{2+} oxidation state in ZnO. There are no secondary zinc phases or metallic zinc if there are no other Zn-related peaks.

The O 1s spectrum is asymmetric and can be deconvoluted into multiple components as displayed in Figure 9c. The main peak located at around 529.8–530.2 eV is attributed to lattice oxygen O^{2-} bonded to Zn^{2+} in the Wz ZnO structure. A secondary peak appearing at higher binding energy (~531.2–531.8 eV) is commonly associated with oxygen vacancies, oxygen-deficient regions, or surface hydroxyl groups.

The presence of this higher-energy O 1s component indicates a significant concentration of oxygen vacancies in the ZnO thin films. These intrinsic defects act as shallow donor states, which is consistent with the n-type conductivity of ZnO and correlates well with the bluish-green PL emission observed at 485 nm.

Correlation With Optical and Structural Properties

The XPS results confirm that the spray pyrolysis process produces stoichiometric ZnO with controlled defect states. The identified oxygen vacancies play a crucial role in enhancing optical emission and charge transport properties. These findings are in good agreement with UV–Vis–IR and PL analyses, as well as with DFT-GGA calculations, which predict stable Wz ZnO with defect-induced electronic states.

Overall, XPS analysis validates the chemical integrity of the ZnO thin films and highlights the role of oxygen-related defects, reinforcing their suitability for optoelectronic, sensing, and energy-related applications.

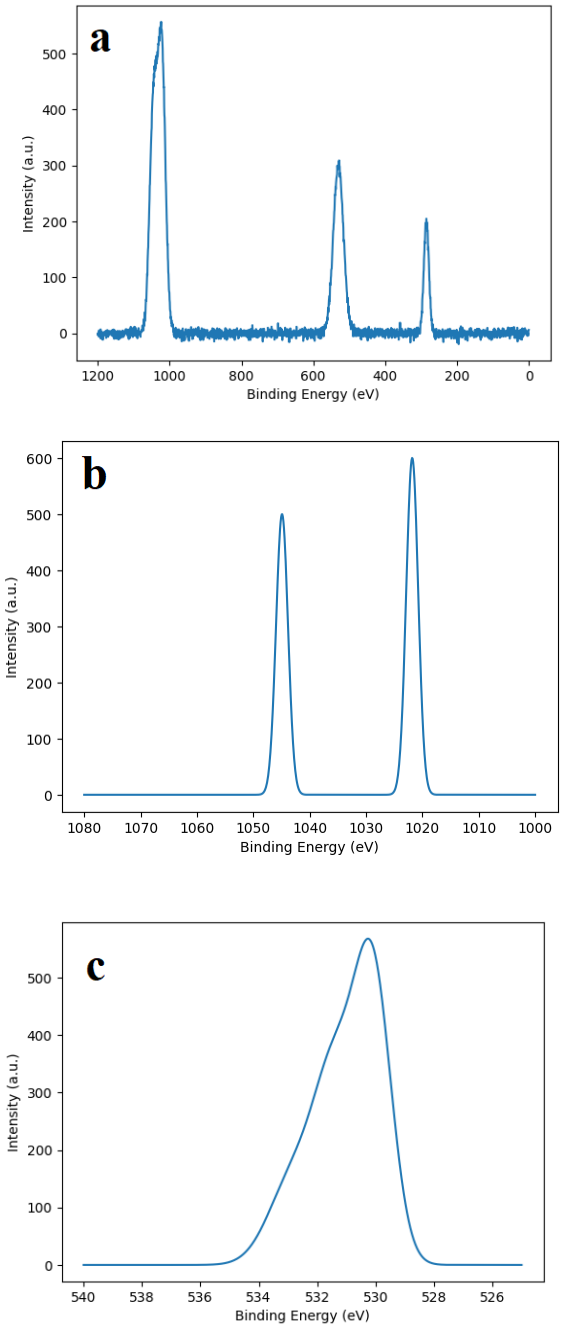


Figure 9: XPS studies of ZnO material (a) spectrum of ZnO within 0-1200 eV, (b) Zn 2p plots and (c) O 1s spectrum.

O 1s peak XPS spectra of ZnO films formed in pure Ar ambient at 300 °C and O2/Ar mixture ambient at 300 °C before and after etching by Ar+ for 10 minutes are studied previously [29]. It is reported that the carbon C 1s peak (284.6 eV) was used as a reference to calibrate the binding energies. The O1s peak of the ZnO films deposited in the pure Ar and O2/Ar mixture ambient at 300 ◦C are studied. So, peak profile of O 1s is decomposed into three parts at 530.4 eV, 531 eV and 532.4 eV as reported by Chen *et al.* [29]. More details are given for the O^{2-} from the Wz structure of the ZnO lattice which is responsible for the minimal-energy peak at 530.4 eV [29], however the O^{2-} in the oxygen-deficient area of the ZnO matrix is responsible for the peak at 531 eV [29].

Computational Studies

It is necessary to comprehend both the electronic structure and how GGA influences it in order to explain and interpret the energy vs. density of states (DOS) plot for ZnO generated using the GGA within DFT.

Figure 10 shows the variation of energy with volume of ZnO materials for its three well-known crystalline structures Rocksalt (R_s), Wz and Z_B.

The graph indicates three curves of E-V plots, each corresponding to a different DFT treatment of ZnO. For all crystalline structure the E-V plot is a parabolic shape recording a minimum.

This profile can be explained by the V_0 related to the lowest value of total energy of ZnO crystal. This fact is the most stable atomic arrangement. Further, V_0 values are recorded at 136.98, 166.2 and 334 $Bohr^3$ for ZnO Rs, ZnO Z_B and ZnO Wz respectively.

The curve of Rs shows a low energy indicating a stability of the ZnO crystal while Wz structure records a larger V_0. This fact is common with GGA method because overvalues the lattice constants.

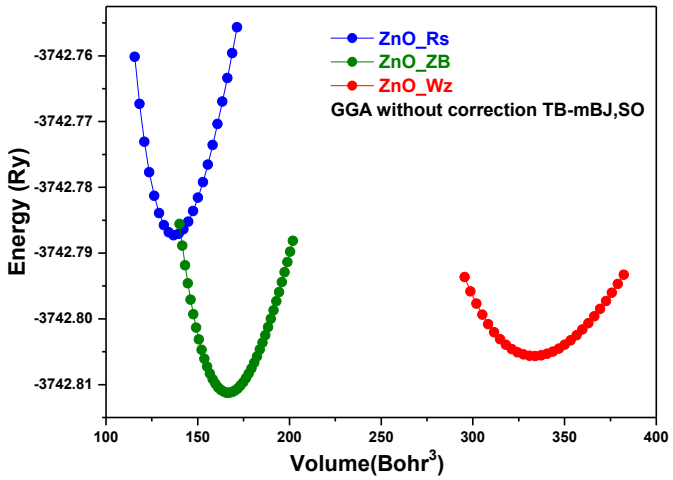


Figure 10: Energy versus volume of ZnO material estimated by GGA approximation for three crystalline structures Rs, ZB and Wz

In this Figure, each point of the curve is related to E energy which the total energy of the crystal when its volume is V. As indicated in Figure 10, three curves are displayed corresponding to three crystalline structures of ZnO the Rs, the Wz and the Z_B. Curve of ZnO displays two regions, one for small volumes (V<<) and another for large volumes (V>>).

The E-V plotting, a parabolic curve, looks like $E\alpha V^2$: in V<< range, the lattice is compressed while V>> the lattice is enlarged. As depicted in Figure 10. Where, 1 Bohr≈ 5.291×10^{-11} m and Rydberg, 1Ry =13.606 eV. Near minimum, E is expressed as [30],

$$E(V) = E_0 + \frac{1}{2}B_0V_0\left(\frac{V-V_0}{V_0}\right)^2 \quad (4)$$

Where E_0 is energy of the lattice at equilibrium, B_0 is bulk modulus

The DOS indicates the number of electronic states accessible to electrons at each energy level in ZnO.

-The curve x-axis represents energy (usually in eV, relative to the Fermi level E_F)

-The curve y-axis represents DOS (number of states per eV per unit cell)

There are usually two DOS types:

-The sum of all atoms and orbitals is known as total DOS (TDOS).

-Contribution from particular atoms or orbitals (such as Zn-3d and O-2p) is known as partial DOS (PDOS). As well known, ZnO is an extensive band gap semiconductor having two bands

One is lower dominated by O-2p orbitals is valence band (VB) and a second is higher from

Zn-4s orbitals is called C_B. While Zn-3d states are found deeper below the VB (usually between –7 and –6 eV), influencing VB shape *via* hybridization.

Evaluation of band-gap problem and finite-size effect correction techniques in supercell defect computations for ZnO and GaAs case studies are earlier investigated [31].

ZnO nanoparticle modeling, optimization, and band gap estimation using first principles are previously mentioned [32].

Figure 11 depicts the Gaussian fit of E-V curve evaluated by GGA approximation for Rs (a), ZB (b) and Wz (c) crystalline structures of ZnO material.

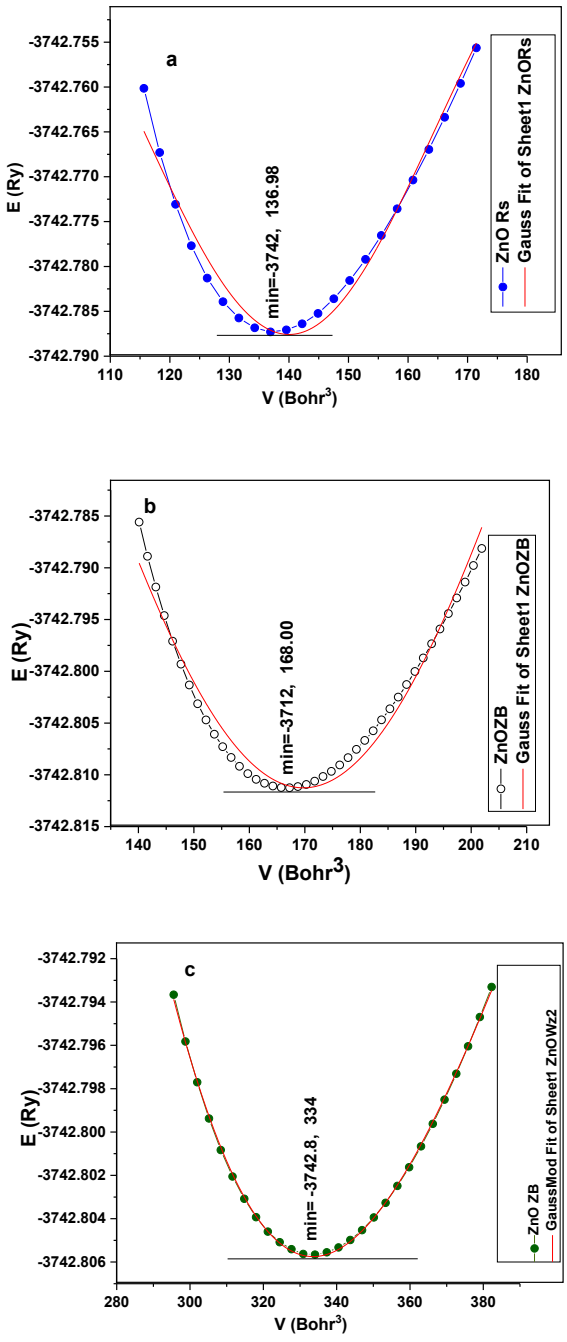


Figure 11: Gaussian fit of E-V curve evaluated by GGA approximation for Rs (a), Z_B (b) and Wz (c) ZnO material. Minimum is displayed

In order to interpret DOS plot, as shown in Figure 12, we can assume this explanation: Band gap (E_g) corresponds to area comprised within VB-CB. In the case of ZnO, experimentally Eg is around 3.3-3.5 eV. But, GGA undervalues Eg at ~0.7–1.0 eV. Because GGA fails to properly treat the exchange–correlation potential for localized d-electrons (Zn-3d), leading to an undervalued C_B energy. For the VB which is mainly composed of O-2p states, interaction (hybridization) between O-2p and Zn-3d at lower energies might occurs.

Additionally, Zn-4s orbitals dominate the C_B. As a result, DOS progressively rises with energy over the C_B minimum (C_{BM}). E_F is around mid-gap for an intrinsic semiconductor. If the plot exhibits EF closer to the C_B, involves a n-type behavior, which is typical for ZnO because of oxygen vacancies. Parameters extracted from the Gaussian fit according to GGA approximation are tabulated in Table 5.

Table 5: Parameters extracted from the Gaussian fit according to GGA approximation

	Minimum (Ry)	y0 (Ry)	xc (Bohr ^{3})	w FWMH	A (a.u.)
ZnO Rs	-3742.78	-3742.73	140.00	48.52	-3.47
ZnO Wz	3742.77	-3742.75	336.50	119.10	-7.57
ZnO Z_B	-3742.69	-3742.67	258.09	55.46	-46.00

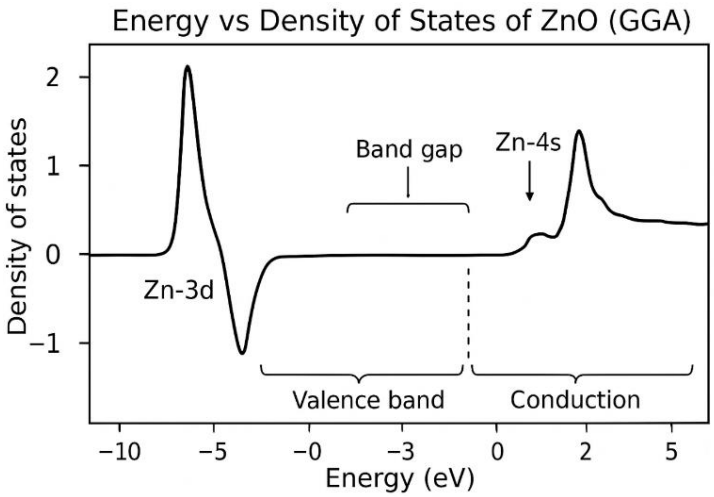


Figure 12: The DOS profile of ZnO material [33]

The Zn-3d, O-2p valence and Zn-4s conduction zones are displayed in Figure also the band gap which separate VB and CB is drawn. Table 6 is listed to interpret and summarize the DOS plot.

Table 6: The energy areas, dominant states, physical interpretation, GGA method

Area	Dominant states	Physical interpretation	In GGA
Deep valence around -7eV	Zn-3d	Zn orbitals are localized	Too shallow of a position (GGA restriction) [33]
Upper valence -5eV to 0 eV	O-2p	Bonding states	Width is 4-5 eV [34]
Band gap around 1 eV according to GGA	-	Forbidden area	Undervalued [35]
CB > 0 eV	Zn-4s	Anti-bonding states	Slight increase in DOS [36]

Conclusions

ZnO thin films were successfully deposited on glass substrates using a low-cost and simple spray pyrolysis technique. UV–Vis–IR spectroscopy confirms that the films exhibit high optical transparency and a wide band gap, consistent with semiconducting behavior. The observed bluish-green emission at 485 nm is attributed to oxygen vacancies, which act as shallow donor levels and facilitate electron transitions accompanied by photon emission. Morphological analysis indicates that the ZnO film is uniform and nanostructured, with regularly distributed grains and no detectable voids. This smooth and compact surface morphology makes the films suitable for applications such as dye-sensitized solar cells, piezoelectric devices, and gas sensors. From the structural analysis, the Rs curve shows low energy, indicating good crystal stability of ZnO. The Wz structure exhibits a larger V_0, which is a known effect of the GGA approximation, as it tends to overestimate lattice parameters. The most reliable structural optimization corresponds to the lowest total energy, consistent with the inherent stability of the Wz phase of ZnO. Overall, the comparison of results highlights the influence of computational approximations on the predicted structural properties of ZnO.

Conflict of Interest

The authors declared no conflict of interest.

Data Availability Statement

All data are available upon request.

Acknowledgements

Our work is included in PRFU project under contract number B00L02UN310220220001 Oran University of Sciences and Technology USTO-MB, A big Thank you to any student/researcher who followed my advice, believed in my advice and contributed in order to improve the FDFCA RG works.

ORCID

Mostefa Benhaliliba
https://orcid.org/0000-0001-6507-3663

References

[1] Benouis, C.E., Benhaliliba, M., Juarez, A.S., Aida, M.S., Chami, F., Yakuphanoglu, F. The effect of indium doping on structural, electrical conductivity, photoconductivity and density of states properties of ZnO films. *Journal of Alloys and Compounds*, **2010**, 490(1-2), 62-67.
[2] Benhaliliba, M. ZnO a multifunctional material: Physical properties, spectroscopic ellipsometry and surface examination. *Optik* , **2021**, 241, 167197.
[3] Benhaliliba, M., Benouis, C.E., Aida, M.S., Yakuphanoglu, F., Sanchez Juarez, A. Indium and aluminium-doped ZnO thin films deposited onto FTO substrates: Nanostructure, optical, photoluminescence and electrical properties. *Sol-Gel Science and Technology*, **2010**, 55, 335–342.
[4] Nur-E-Alam, M., Kar Yap, B., Aminul Islam, M., Chou Yong, T., Seih Kiong, T., Yoong Chan, K., Soon How Thien, G., Khairul Basher, M., Tamam, N., Khandaker, M.U. Spin-coated mg-doped ZnO thin films as electron transport layers for efficient and stable perovskite solar cells. *Scientific Reports*, **2025**, 15, 36618.
[5] Chen, Z., Wang, J., Wu, H., Yang, J., Wang, Y., Zhang, J., Bao, Q., Wang, M., Ma, Z., Tress, W., Tang, Z. A transparent electrode based on solution-processed ZnO for organic optoelectronic devices. *Nature Communications*, **2022**, 13, 4387.
[6] Wang, C., Xie, J., Chang, X., Zheng, W., Zhang, J., Liu, X. ZnO single nanowire gas sensor: A platform to investigate the sensitization of Pt. *Chemical Engineering Journal*, **2023**, 473, 145481.
[7] Wei, T., Zhang, N., Ji, Y., Zhang, J., Zhu, Y., Yi, T. Nanosized zinc oxides-based materials for electrochemical energy storage and conversion: Batteries and supercapacitors. *Chinese Chemical Letters*, **2022**, 33(2), 714-729.
[8] Fortunato, E., Barquinha, P., Pimentel, A., Goncalves, A., Marques, A., Pereira, L., Martins, R. Recent advances in ZnO transparent thin film transistors. *Thin Solid Films*, **2005**, 487, 205–211.
[9] Constantinoiu, I., Viespe, C. ZnO metal oxide semiconductor in surface acoustic wave sensors: A review. *Sensors*, **2020**, 20, 5118.
[10] Baig, A., Siddique, M., Panchal, S. A review of visible-light-active zinc oxide photocatalysts for environmental application. *Catalysts*, **2025**, 15(2), 100.

[11] El-Fallal, A.A., Elfayoumy, R.A., El-Zahed, M.M. Antibacterial activity of biosynthesized zinc oxide nanoparticles using Kombucha extract. *SN Applied Sciences*, **2023**, 5, 332.
[12] Puspasari, V., Ridhova, A., Hermawan, A., Amal, M.I., Khan, M.M. ZnO-based antimicrobial coatings for biomedical applications. *BioSystems*, **2022**, 45(9), 1421–1445.
[13] Mokhtari, H., Benhaliliba, M., Boukhachem, A., Aida, M.S., Ocak, Y.S. Nanostructured device based on coated ZnO layer as a window in solar cell applications. *Materials Science-Poland*, **2017**, 36(4), 570-583.
[14] Shafi, M.A., Bouich, A., Fradi, K., Guaita, J.M., Khan, L., Mari, B. Effect of deposition cycles on the properties of ZnO thin films deposited by spin coating method for CZTS-based solar cells. *Optik*, **2022**, 258, 168854.
[15] Abdallah, B., Zetoun, W., Tello, A. Deposition of ZnO thin films with different powers using RF magnetron sputtering method: Structural, electrical and optical study. *Heliyon*, **2024**, 10(6), e27606.
[16] Güçbilmez, Ö.B., Tekin, S., Acar, S. Tuning of linear and nonlinear optical properties of ZnO nanostructured films by varying the alkanolamine stabilizers. *Journal of Alloys and Compounds*, **2025**, 1037, 182606.
[17] Shahrokhi, M., Leonard, C. Tuning the band gap and optical spectra of silicon-doped graphene: Many-body effects and excitonic states. *Journal of Alloys and Compounds*, **2017**, 693, 1185-1196.
[18] Panish, M.B., Casey, H.C. Temperature dependence of the energy gap in GaAs and GaP. *Journal of Applied Physics*, **1969**, 40, 163–167.
[19] Monemar, B. Fundamental energy gap of GaN from photoluminescence excitation spectra. *Physical Review B*, **1974**, 10, 676.
[20] Perez, J.P., Pedrero, E., Lazcano, Y.R., Rodríguez, J.A. Morphology, microstructure and optical properties of nanostructured ultrasonically sprayed ZnO thin films with different thicknesses. *Optical Materials*, **2025**, 162, 116899.
[21] Benouis, C.E., Benhaliliba, M., Mouffak, Z., Avila-Garcia, A., Tiburcio-Silver, A., Ortega Lopez, M., Romano Trujillo, R., Ocak, Y.S. The low resistive and transparent Al-doped SnO_2 films: p-Type conductivity, nanostructures and photoluminescence. *Journal of Alloys and Compounds*, **2014**, 603, 213-223.
[22] Li, Z., Chen, Y., Meng, X., Zhang, B., Fan, G., Zhou, Y., Zhang, S., Hu, R., Xu, F. Band engineering and enhanced photoluminescence via Al donor doping in ZnO thin films. *Physica B: Condensed Matter*, **2025**, 717, 417788.
[23] Benhaliliba, M., Benouis, C.E., Aldemir, D.A. The presence of C/ω-V and G/ω-V peaks profile of Ag/SnO_2/n-Si/Au MOS junction for capacitor applications. *Physica B: Condensed Matter*, **2019**, 572, 175–183.
[24] Kaviyarasu, K., Magdalane, C.M., Kanimozhi, K., Kennedy, J., Siddhardha, B., Subba Reddy, E., Rotte, N.K., Sharma, C.S., Thema, F.T., Letsholathebe, D., Mola, G.T., Maaza, M. Elucidation of photocatalysis, photoluminescence and antibacterial studies of ZnO thin films by spin coating method. *Journal of Photochemistry and Photobiology B: Biology*, **2017**, 173, 466-475.
[25] Ghosh, J., Ghosh, R., Giri, P.K. Tuning the visible photoluminescence in Al doped ZnO thin film and its application in label-free glucose detection. *Sensors and Actuators B: Chemical*, **2018**, 254, 681-689.
[26] Lim, J., Shin, K., Kim, H.W., Lee, C. Photoluminescence Studies of ZnO thin films grown by atomic layer epitaxy. *Journal of Luminescence*, **2004**, 109, 181-185.
[27] Bappy, N.F., Subramani, S. A comprehensive review on Mg-doped ZnO thin film and nanostructure: Properties and applications. *Materials Science and Engineering: B*, **2025**, 318, 118251.
[28] Lin, S.S., Huang, J.L. Effect of thickness on the structural and optical properties of ZnO films by r.f. magnetron sputtering. *Surface & Coatings Technology*, **2004**, 185, 222–227.
[29] Chen, T., Liu, S.Y., Xie, Q., Detavernier, C., Van Meirhaeghe, R.L., Qu, X.P. The effects of deposition temperature and ambient on the physical and electrical performance of DC-sputtered n-ZnO/p-Si heterojunction. *Applied Physics A*, **2010**, 98, 357–365.
[30] Rana, M.R., Islam, S., Hoque, K., Biswas, G.G., Hossain, M.E., Naqib, S.H., Ali, M.A. DFT prediction of the stability and physical properties of M2GaB (M = Sc, V, Nb, Ta). *Journal of Materials Research and Technology*, **2023**, 24, 7795-7815.
[31] Lany, S., Zunger, A. Assessment of correction methods for the band-gap problem and for finite-size effects in supercell defect calculations: Case studies for ZnO and GaAs. *Physical Review B*, **2008**, 78, 235104.
[32] Harun, K., Mansor, N., Ahmad, Z.A., Mohamad, A.A. Electronic properties of ZnO nanoparticles synthesized by Solgel method: A LDA+U calculation and experimental study. *Procedia Chemistry*, **2016**, 19, 125–132.
[33] Oba, F., Choi, M., Togo, A., Tanaka, I. Point defects in ZnO: An approach from first principles. *Science and Technology of Advanced Materials*, **2011**, 12(3), 034302.
[34] Vogel, D., Krüger, P., Pollmann, J. Self-interaction and relaxation-corrected pseudopotentials for II-VI semiconductors. *Physical Review B*, **1996**, 54, 5495.
[35] Harun, K., Salleh, N.A., Deghfel, B., Yaakob, M.K., Mohamad, A.A. DFT + U calculations for electronic,

structural, and optical properties of ZnO wurtzite structure: A review. *Results in Physics*, **2020**, 16, 102829.

[36] Szakacs, C.E., Merschrod, E.F.S., Poduska, K.M. Structural features that stabilize zno clusters: An electronic structure approach. *Computation*, **2013**, 1(1), 16-26.

Copyright © 2026 by SPC (Sami Publishing Company) + is an open access article distributed under the Creative Commons Attribution License (CC BY), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work is properly cited.